PROJECT SCOPE & STRATEGIC PROPOSAL

UK Train Rides Data Analysis & Forecasting System

Unlocking Operational Insights, Demand Dynamics, and Revenue Optimization via Power BI, Power Query & DAX

A comprehensive business intelligence initiative designed to transform national railway transaction records into actionable strategic intelligence. This proposal details the end-to-end analytical architecture, implementation methodology, governance framework, and task allocation for the DEPI Project Initiative.

Project Name:	UK Train Rides Analytics
Project Lead:	Nada Atia Abd Rabbo Mousa
Core Team:	Zeinab M. Sheta, Khalid W. Mohamed, Doaa A. Awaden, Abdullah A. Al-Khazraji
Primary Stack:	Power BI Desktop / Service, Power Query ETL, DAX Engine
Target Completion:	4 Weeks Execution Plan
Document Ref:	DEPI-PRJ-UKTR-2026-V1.0

EXECUTIVE NOTICE

This document constitutes the formal project plan and operational baseline. All scope specifications, deliverable schedules, and resource assignments outlined herein have been aligned with stakeholder directives and repository specifications.

1. Table of Contents & Executive Summary

Table of Contents

1. Executive Summary & Table of Contents — Page 2

2. Background & Strategic Context — Page 3

3. Project Objectives & Analytical Focus — Page 4

4. Technology Stack & Technical Architecture — Page 5

5. Data Transformation & Power Query Methodology — Page 6

6. DAX Modeling & Predictive Forecasting Strategy — Page 7

7. Dashboard Architecture & User Experience Design — Page 8

8. Deliverables Schedule & Implementation Roadmap — Page 9

9. Team Governance, Task Allocation & Sign-Off — Page 10

Executive Summary

The UK passenger rail network is a multi-faceted transportation ecosystem serving millions of commuters and leisure travelers daily. Optimizing service efficiency, ticket pricing strategies, and route capacities requires precise, data-driven decision-making. The **UK Train Rides Data Analysis Project** is established to convert complex transactional journey data into interactive, intuitive intelligence dashboards utilizing Microsoft Power BI, Power Query, and Advanced DAX (Data Analysis Expressions).

This proposal outlines the foundational roadmap, technical strategy, and task distribution for our specialized five-member analysis team over a structured 4-week project lifecycle. Key outcomes include automated data integration, predictive ride volume forecasting, daily revenue projection for upcoming months, and passenger segmentation across ticket classes and travel corridors.

4 Weeks	5 Members	100% Data	Daily & Monthly
PROJECT TIMELINE	SPECIALIZED TEAM	AUTOMATED ETL	REVENUE FORECASTS

2. Background & Strategic Context

Modern rail operations generate massive amounts of transactional data across ticket purchasing systems, station check-ins, delayed journey logs, and route scheduling platforms. However, raw data remains underutilized without clean, structured pipelines and intuitive business intelligence platforms.

Based on the project governance framework established in the official project repository (**NadaAtia688/DEPI-Project-**), our initiative responds directly to the need for clear executive analytics. Decision-makers require actionable clarity on passenger volume trends, peak congestion periods, revenue performance by ticket tier (First Class vs. Standard), and reliable forecasts to allocate operational resources effectively for the upcoming months.

Data Environment & Analytical Challenges

Data Fragmentation & Quality

Raw transaction logs contain inconsistent date-time stamps, missing journey delay reasons, and unstandardized station names that require comprehensive scrubbing in Power Query before modeling.

Revenue & Class Variance

Ticket prices fluctuate significantly across Advance, Off-Peak, and Anytime purchases. Understanding price elasticity across ticket classes is vital for financial forecasting.

Demand Peak Volatility

Passenger density varies drastically between weekday commuter rushes and weekend leisure travel, requiring precise daily and hour-by-hour trend modeling.

Forward Projection Needs

Management needs more than retrospective analysis; they require a predictive model estimating ride counts and daily revenue for the next 30 days to optimize staffing and rolling stock.

> **Strategic Alignment Notice**
> This project direct aligns with organizational goals by establishing a single source of truth for rail performance metrics, reducing reporting latency from days to near real-time interactivity.

3. Project Objectives & Analytical Focus

The primary objective of the UK Train Rides Data Analysis Project is to build a robust, scalable, and interactive Power BI analytical ecosystem. The project is structured around answering core operational and financial questions vital for rail executive management.

1. Clean & Standardize Rail Data

Establish an automated Data Preprocessing pipeline using Power Query to ingest, clean, handle null values, and standardize raw journey datasets into an optimized star schema.

2. Passenger Demand Forecasting

Predict total ride volume for the next month, analyzing daily passenger distributions, weekday vs. weekend patterns, and peak travel window frequencies.

3. Daily Revenue Projections

Develop dynamic DAX measures to forecast daily revenue for the next month, identifying top revenue-generating corridors and fare structure efficiencies.

4. Ticket Class & Breakdown Analysis

Evaluate passenger demand across distinct ticket classes (First Class, Standard) and ticket types (Advance, Anytime, Off-Peak) to optimize pricing models.

Core Key Performance Indicators (KPIs)

- **Total Rides & Passenger Throughput:** Historical count and forecasted volume for upcoming 30-day windows.
- **Total & Forecasted Revenue:** Monitored daily, weekly, and monthly across all regional routes.
- **On-Time Performance Rate (%):** Journeys arriving within scheduled time windows vs. delayed/canceled rides.
- **Average Ticket Yield:** Average revenue earned per passenger per kilometer across ticket classes.
- **Peak Utilization Ratio:** Hourly passenger concentration during morning and evening rush hours.

4. Technology Stack & Technical Architecture

To ensure optimal performance, scalability, and seamless user adoption, the project leverages the complete Microsoft Power BI suite. Each component plays a distinct, vital role in the data processing and analytics pipeline.

Power Query (ETL Engine)

Serves as the primary Extract, Transform, and Load (ETL) layer. Responsible for connecting to raw CSV/database sources, performing data cleansing, type conversions, conditional column creation, handling missing journey records, and merging station reference tables.

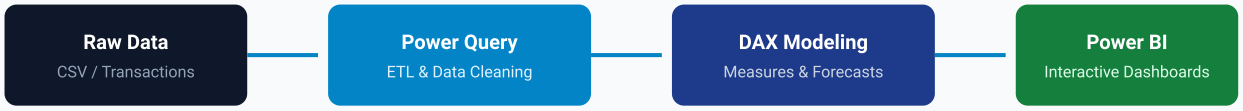
DAX - Data Analysis Expressions (Analytical Engine)

Powers the data model's intelligence. DAX is used to write time-intelligence measures, dynamic dynamic aggregations, moving averages for demand forecasting, ticket class distribution ratios, and month-over-month revenue comparison formulas.

Power BI Desktop & Service (Visualization Platform)

Provides the front-end user experience. Enables the creation of interactive, drag-and-drop visuals, dynamic slicers (by date, station, ticket class), custom tooltips, and seamless publishing to Power BI Service for stakeholder collaboration.

End-to-End Data Pipeline Visual



5. Data Transformation & Power Query Methodology

Data preprocessing forms the backbone of reliable analytics. During Week 1, the team will focus on extracting raw UK rail dataset files and transforming them into clean, structured tables suitable for star-schema modeling in Power BI.

Key Power Query Transformation Steps

Transformation Step	Action Items	Expected Outcome
Data Profiling	Examine column quality, value distribution, and identify missing records.	Baseline report on missing data percentages and anomaly counts.
Date & Time Parsing	Split timestamps into Date, Time, Hour, and Day of Week columns.	Enables granular peak-hour and day-type time intelligence.
Text Standardization	Trim spaces, capitalize station names, and standardize ticket categories.	Eliminates duplicate categories caused by minor string variations.
Null Value Handling	Impute or isolate unrecorded delay reasons and missing price fields.	Ensures accurate financial calculations without dropping key rides.
Custom Columns	Create conditional logic for Journey Status (On-Time, Delayed, Canceled).	Simplified operational reporting dimensions for visuals.
Schema Design	Separate Fact Table (Transactions) from Dimension Tables (Stations, Dates).	Optimized Star-Schema for high DAX query execution speed.

> **Week 1 Data Preprocessing Deliverable Notebook**
> In addition to the cleaned dataset within Power BI, the team will deliver a structured **Data Preprocessing Notebook** documenting every Applied Step in Power Query, ensuring full repeatability and auditability.

6. DAX Modeling & Predictive Forecasting Strategy

Week 2 focuses on formulating critical business questions and constructing advanced DAX calculations. By leveraging time-intelligence and dynamic forecasting logic, Power BI will project next month's passenger demand and daily revenue streams.

Key Analytical DAX Measures

1. Forecasted Passenger Demand (Next Month)

Calculates daily ride volume expectations by applying a weighted 3-month moving average combined with weekday growth factors.

```
Forecasted_Rides = CALCULATE(AVERAGEX(VALUES('Calendar'[Date]), [Total Rides]),
DATEADD('Calendar'[Date], -1, MONTH)) * [Growth_Factor]
```

2. Daily Revenue Projection

Projects daily revenue for each day of the upcoming month based on historical ticket class distribution and average yield per class.

```
Daily_Forecasted_Revenue = SUMX(VALUES('Ticket Class'[Class]), [Forecasted_Rides_By_Class]
* [Avg_Ticket_Price])
```

3. Ticket Class Demand Breakdown Ratio

Evaluates proportional demand share across First Class and Standard, allowing dynamic filtering across peak and off-peak journey slots.

```
Class_Demand_Share = DIVIDE([Rides by Ticket Class], [Total Rides], 0)
```

7. Dashboard Architecture & User Experience Design

During Week 3, the backend data models and DAX calculations will be transformed into an executive-ready, highly intuitive Power BI Dashboard. The user interface will emphasize visual clarity, minimal cognitive load, and effortless navigation.

Planned Dashboard Layout Structure

Executive Summary Tab

High-level KPI cards showing Total Rides, Total Revenue, On-Time %, and Top Corridor. Includes revenue trend line charts and regional route maps.

Predictive Demand & Revenue Tab

Dedicated forecasting view presenting next month's projected ride counts per day, daily revenue expectations, and ticket class demand splits.

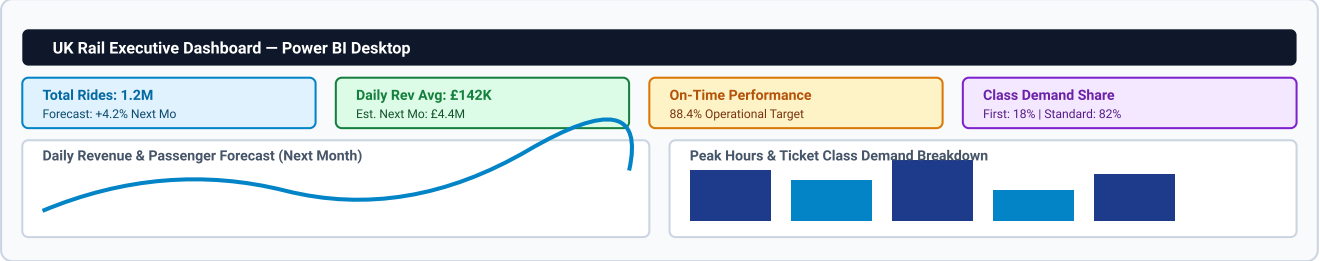
Punctuality & Operations Tab

Detailed analysis of journey delays, peak congestion hours, delay root causes, and station-by-station traffic volume.

Ticket & Fare Analytics Tab

Deep dive into ticket types (Advance, Anytime, Off-Peak), purchase channels, revenue contribution by ticket class, and discount usage.

Visual Dashboard Wireframe Preview



8. Deliverables Schedule & Implementation Roadmap

The project follows a rigorous 4-week execution lifecycle. Each phase includes specific, verifiable deliverables that ensure transparency, quality control, and alignment with stakeholder objectives.

Phase	Focus Area	Key Tasks	Phase Deliverables
Week 1	Data Cleaning & Preprocessing	Ingest raw data into Power Query, profile columns, clean nulls, format timestamps, and build star schema.	• Cleaned Dataset • Data Preprocessing Notebook
Week 2	Analysis Questions Phase	Define decision-maker questions, author DAX forecasting measures for ride counts and daily revenue, analyze ticket class demand.	• Verified Question Set • DAX Analytical Calculations
Week 3	Dashboard Phase	Construct interactive Power BI dashboard, design UI tabs, add dynamic slicers, format visual cross-filtering.	• Interactive Power BI Dashboard (.pbix file)
Week 4	Final Presentation	Synthesize findings into executive presentation, document model deployment, present final insights to stakeholders.	• Final Project Report • Executive Presentation

Quality Assurance & Review Gates

At the end of each week, the project team will conduct an internal milestone review to validate deliverables against repository guidelines before advancing to the subsequent phase.

9. Team Governance, Task Allocation & Sign-Off

To maximize efficiency and leverage individual strengths, tasks are distributed across our five-member project team under the leadership of Team Lead Nada Atia. All collaboration, code, and document assets are maintained in the central GitHub repository (**DEPI-Project-**).

Responsibility Assignment Matrix (RACI)

Team Member	Primary Role	Assigned Responsibilities & Focus Areas
Nada Atia Abd Rabbo Mousa	Team Lead (TL) & Project Manager	Overall project governance, milestone tracking, stakeholder communications, final report synthesis, and Power BI dashboard architecture integration.
Zeinab Mohamed Aboelftoh Sheta	Lead Data Engineer	Power Query ETL execution, data profiling, timestamp parsing, missing value imputation, and Data Preprocessing Notebook authoring.
Khalid Walid Mohamed	DAX & Modeling Specialist	Star-schema data modeling, authoring DAX time-intelligence formulas, dynamic revenue forecasting, and passenger volume calculations.
Doaa Ahmed Abd Elaziz Awaden	BI Visuals Developer	Power BI dashboard UI layout design, KPI visual design, ticket class distribution charts, and user experience interactivity.
Abdullah Al-Khazraji M. Al-Khazraji	Quality & Presentation Analyst	Data validation, testing DAX measures for precision, drafting the final presentation deck, and supporting business insight extraction.

Closing & Stakeholder Feedback Invitation

This project proposal establishes the baseline for the UK Train Rides Data Analysis initiative. We invite project stakeholders and external reviewers to review this document and provide feedback or sign-off to initiate Week 1 operations.

Contact: Nada Atia (Team Lead) | Repository: NadaAtia688/DEPI-Project-